

An Educational Derivation of the Dodecahedron Dihedral Angle

Folding a three-pentagon patch until two vertices coincide

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Main result. If two congruent pentagon faces are folded out of the original xy -plane by the same physical angle and the moving vertices C and K are required to meet, then the fold angle is

$$\theta = \arccos\left(\frac{1}{\sqrt{5}}\right) = \arctan(2) \approx 63.43494882^\circ.$$

The corresponding *internal dihedral angle* of the regular dodecahedron is

$$\delta = 180^\circ - \theta \approx 116.56505118^\circ.$$

Abstract

This note shows how a local patch of three regular pentagons already contains the dihedral angle of the regular dodecahedron. We first set up a coordinate model, then derive the exact fold angle using Rodrigues' rotation formula and a symmetry condition. We also give a numerical minimization viewpoint, which is often the most practical way to solve related folding problems computationally.

1. Geometric setup

At the shared vertex O , the relevant directions in the planar pentagon patch are the hinge edges OA and OH , together with the moving edges OC and OK . We place O at the origin and normalize lengths so that $|OA| = |OH| = |OC| = |OK| = 1$.

A convenient choice of coordinates is

$$\begin{aligned} A &= (-\cos 36^\circ, \sin 36^\circ, 0), & H &= (\cos 36^\circ, \sin 36^\circ, 0), \\ C &= (-\cos 72^\circ, -\sin 72^\circ, 0), & K &= (\cos 72^\circ, -\sin 72^\circ, 0). \end{aligned}$$

What folds, exactly? The left face is rotated out of the plane about the axis OA . The right face is rotated by the same *physical* fold angle about the axis OH . In a right-hand-rule convention these two rotations carry opposite signs, so we write

$$C_\theta = R_{OA}(\theta) C, \quad K_\theta = R_{OH}(-\theta) K.$$

Because the configuration is mirror-symmetric about the plane $x = 0$, the meeting point must lie in that symmetry plane.

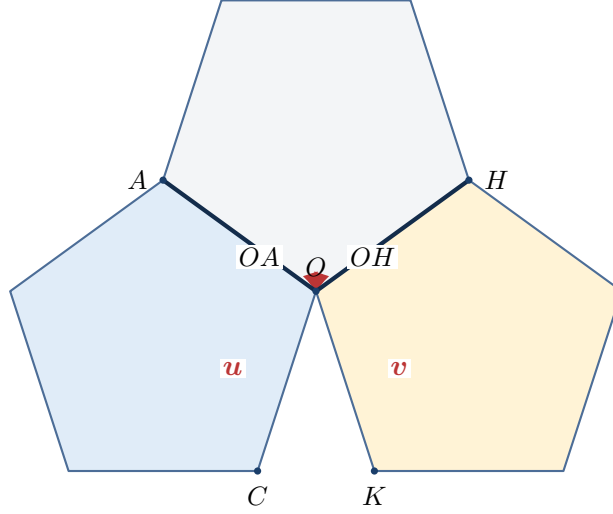


Figure 1: The planar three-pentagon patch. The center face stays in the original plane, and the two side faces fold about the hinge edges OA and OH . Only the moving vertices C and K are labeled.

2. Exact derivation

Let

$$a = \overrightarrow{OA} = (-\cos 36^\circ, \sin 36^\circ, 0), \quad c = \overrightarrow{OC} = (-\cos 72^\circ, -\sin 72^\circ, 0).$$

The Rodrigues formula for rotating a vector c about the unit axis a is

$$R_a(\theta)c = c \cos \theta + (a \times c) \sin \theta + a (a \cdot c)(1 - \cos \theta).$$

We only need the x -coordinate of $C_\theta = R_a(\theta)c$.

Two quick computations are enough:

$$a \cdot c = \cos 36^\circ \cos 72^\circ - \sin 36^\circ \sin 72^\circ = \cos 108^\circ = -\cos 72^\circ,$$

and since a and c lie in the xy -plane,

$$a \times c = (0, 0, \sin 72^\circ).$$

Therefore the cross-product term contributes no x -component, and

$$\begin{aligned} x(C_\theta) &= -\cos 72^\circ \cos \theta + (-\cos 36^\circ)(-\cos 72^\circ)(1 - \cos \theta) \\ &= \cos 72^\circ \cos 36^\circ - \cos 72^\circ(1 + \cos 36^\circ) \cos \theta. \end{aligned}$$

Now use the exact identities

$$\cos 72^\circ \cos 36^\circ = \frac{1}{4}, \quad \cos 72^\circ(1 + \cos 36^\circ) = \frac{\sqrt{5}}{4}.$$

This gives the compact formula

$$x(C_\theta) = \frac{1 - \sqrt{5} \cos \theta}{4}.$$

By symmetry K_θ is the mirror image of C_θ across the plane $x = 0$. Hence C_θ and K_θ meet if and only if $x(C_\theta) = 0$. So

$$\frac{1 - \sqrt{5} \cos \theta}{4} = 0 \quad \implies \quad \cos \theta = \frac{1}{\sqrt{5}}.$$

Thus

$$\theta = \arccos\left(\frac{1}{\sqrt{5}}\right) = \arctan(2) \approx 63.4349882^\circ.$$

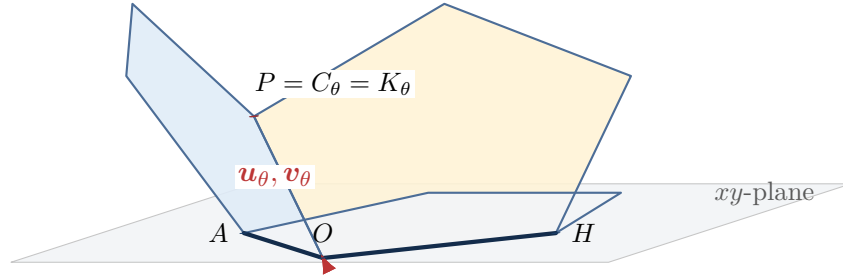


Figure 2: A cleaner 3D view at the meeting angle. The center face stays in the xy -plane while the two side faces rotate about OA and OH until the moving vertices coincide at P .

3. A numerical solution by error minimization

The same answer can be recovered without symbolic manipulation. Define the squared mismatch

$$E(\theta) = \|C_\theta - K_\theta\|^2.$$

This is a nonnegative scalar function of a single variable. The physically correct fold angle is the minimizer of E ; at the true meeting angle the mismatch is exactly zero.

For the present geometry,

$$E(\theta) = \|R_{OA}(\theta)C - R_{OH}(-\theta)K\|^2,$$

and a one-dimensional search over $0^\circ \leq \theta \leq 90^\circ$ gives a unique minimum at

$$\theta \approx 63.43495^\circ.$$

Why this numerical viewpoint matters. The minimization formulation generalizes immediately to more complicated nets, non-regular faces, or situations where an exact symbolic derivation is inconvenient. In practice one can use a simple grid search followed by Brent's method or Newton refinement.

4. From fold angle to dihedral angle

The fold angle θ is the amount each face is rotated out of the original plane. The *internal dihedral angle* δ of the regular dodecahedron is the supplement of that fold angle:

$$\delta = 180^\circ - \theta.$$

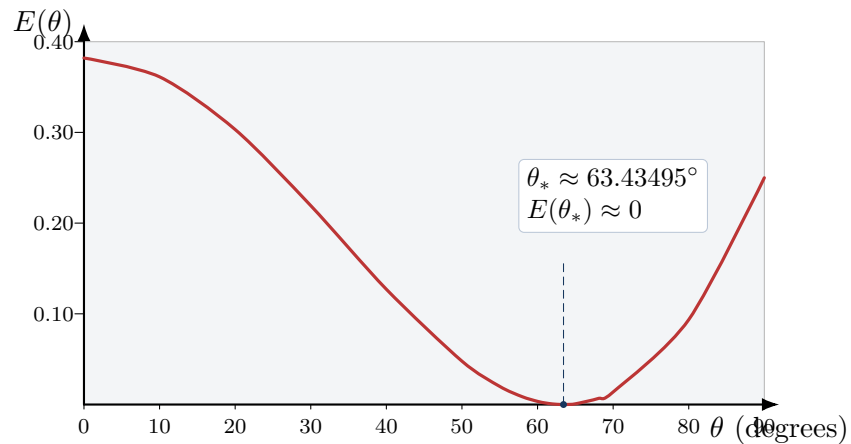


Figure 3: Squared mismatch between the two moving vertices. The minimizer occurs exactly where the two points meet.

Substituting the value above gives

$$\delta = 180^\circ - \arctan(2) \approx 116.56505118^\circ.$$

This is the familiar exact angle for a regular dodecahedron.

Takeaway. The entire dihedral-angle calculation boils down to a local matching condition in a three-pentagon patch. Symmetry reduces the geometry to one scalar equation, and the same geometry also appears naturally as a one-variable minimization problem.